

Modules

Modular Programming

What is a Module?

- Python module is simply a file
- This file helps to organize your Python code
- A module may contain functions

1 Built-in Module (math)

```
import math

# Find G.C.D of two numbers
print ("\n\nG.C.D of 10 and 24 is : ", math.gcd(10, 24))

# Check in what's inside
for Built_Fun in dir(math):
    print (Built_Fun, end=", ")
```

```
G.C.D of 10 and 24 is :  2
__doc__, __loader__, __name__, __package__, __spec__, acos, acosh, asin, asinh,
atan, atan2, atanh, ceil, copysign, cos, cosh, degrees, e, erf, erfc, exp, expm1
, fabs, factorial, floor, fmod, frexp, fsum, gamma, gcd, hypot, inf, isclose, is
finite, isinf, isnan, ldexp, lgamma, log, log10, log1p, log2, modf, nan, pi, pow
, radians, sin, sinh, sqrt, tan, tanh, tau, trunc,
```

2 Sample Program to Test a User Module

```
# Calling entire "NameList" module
import NameList

print ("I'm the main program")

# Calling "getMyName" inside "NameList" module
NameList.getMyName()

# Calling "getMyMiddleName" inside "NameList" module
NameList.getMyMiddleName()
```

Main Program

```
def getMyName():
    print ("Nishan")
    return

def getMyMiddleName():
    print ("Saliya")
    return
```

NameList

3 Specific Function Only!

```
# Calling  "NameList" module but one function only
from NameList import getMyName

print ("I'm the main program")

# Calling "getMyName" inside "NameList" module
# Direct access
getMyName()

# What do you see?
getMyMiddleName()
```

Main Program

```
def getMyName():
    print ("Nishan")
    return

def getMyMiddleName():
    print ("Saliya")
    return
```

NameList

4 Calling a Specific Function - An Alias !!

```
# Calling a single function with a different name
from NameList import getMyName as GN

print ("I'm the main program")

# Calling "getMyName" inside "NameList" module
# Direct access
GN()
```

Main Program

```
def getMyName():
    print ("Nishan")
    return

def getMyMiddleName():
    print ("Saliya")
    return
```

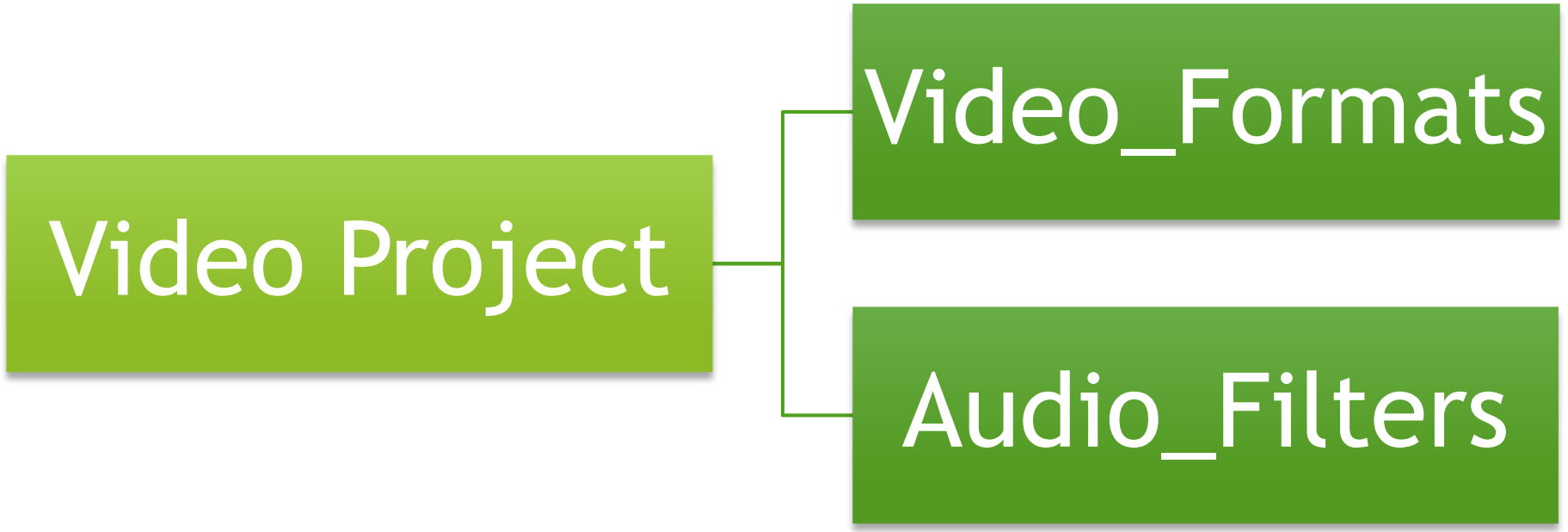
NameList

What is a Package?

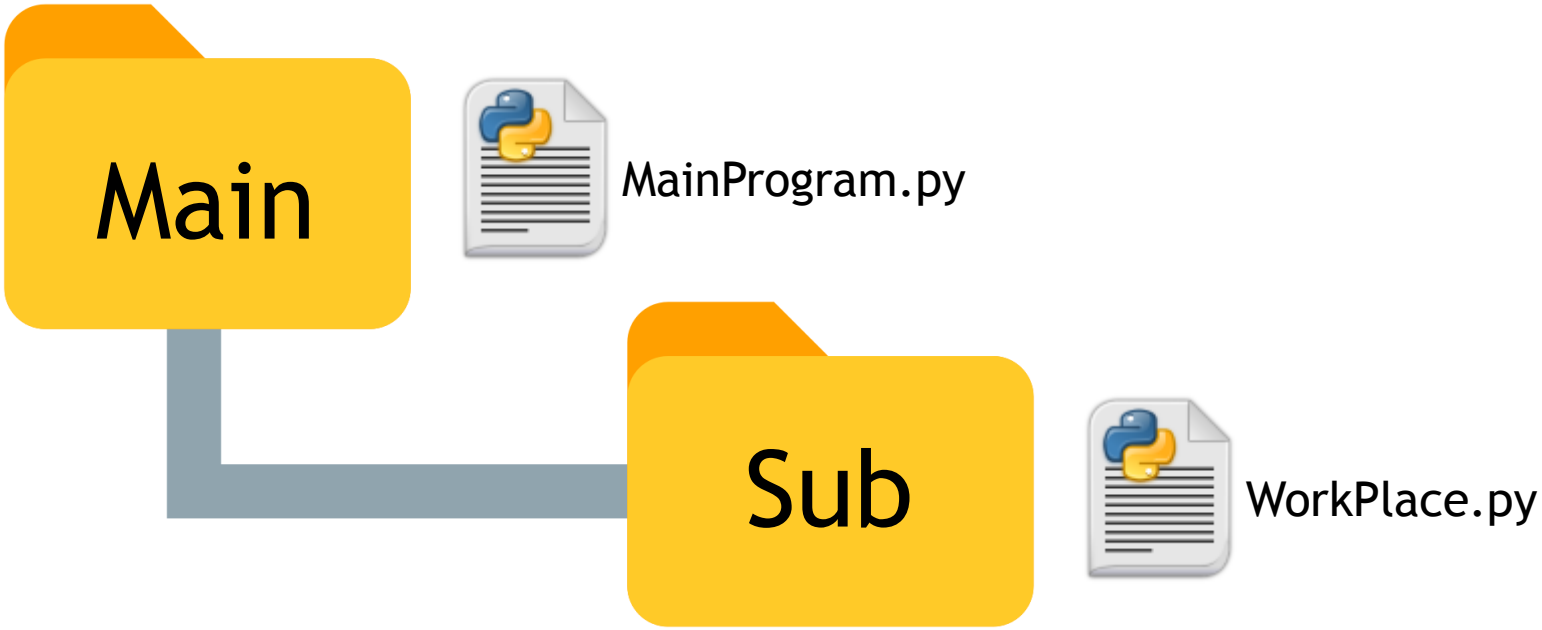
- A package allows your program to be organize
- Even functions can be greatly put in to packages for further organizing them
- They are placed in sub folders



A Sample Package Structure



Your Next Program Structure



5 Testing Packages

```
# Calling from a package
import Sub.WorkPlace

print ("I'm the main program")

# Calling "getShortWorkPlace" inside "WorkPlace" module
Sub.WorkPlace.getShortWorkPlace()
Sub.WorkPlace.getLongWorkPlace()
```

Main Program

```
def getShortWorkPlace():
    print ("IIT")
    return

def getLongWorkPlace():
    print ("Informatics")
    return
```

NameList